MACHINE LEARNING COURSE

Rossmann Store Sales Forecasting

Leveraging Machine Learning to Drive Operational Excellence across 1,115 Rossmann Drug Stores

March 2026 · Olivier ANDRIANOEL – Wassim BELABED – Thomas MAGALLON

AGENDA

Today's Presentation

01
Business Context & Dataset
Why sales forecasting matters for Rossmann operations

02
Data Exploration
Seasonality, promotions, weekday patterns and store dynamics

03
Modelling Approach
Feature engineering, temporal validation and baseline benchmarks

04
Results & Model Comparison
RMSPE vs MAE trade-off, actual vs predicted analysis

05
Error Diagnostics
Residual structure, store-level failures, calendar effects

06
Business Implications
Operational recommendations and path to production

The Commercial Problem We Are Solving

Rossmann: 1,115 European drug stores. Daily revenue forecasting at store level.

Stock Replenishment

Without daily demand visibility, stores over-order or face stockouts, both of which destroy margin.

Labor Scheduling

Scheduling staff without a demand forecast means overtime waste or understaffing peaks.

Revenue Visibility

Finance requires reliable near-term revenue projections for liquidity planning.

Promo Optimization

Promo ROI cannot be measured or optimized without isolating the demand uplift.

OBJECTIVE

Dataset at a Glance

Merged train.csv + store.csv • 2013-01-01 to 2015-07-31

1,017,209
Total Records
After merging train + store

1,115
Unique Stores
Cross-sectional panel data

844,338
Modelling Rows
Open stores, sales > 0

2.5 years
Time Horizon
Jan 2013 – Jul 2015

MODELLING PIPELINE

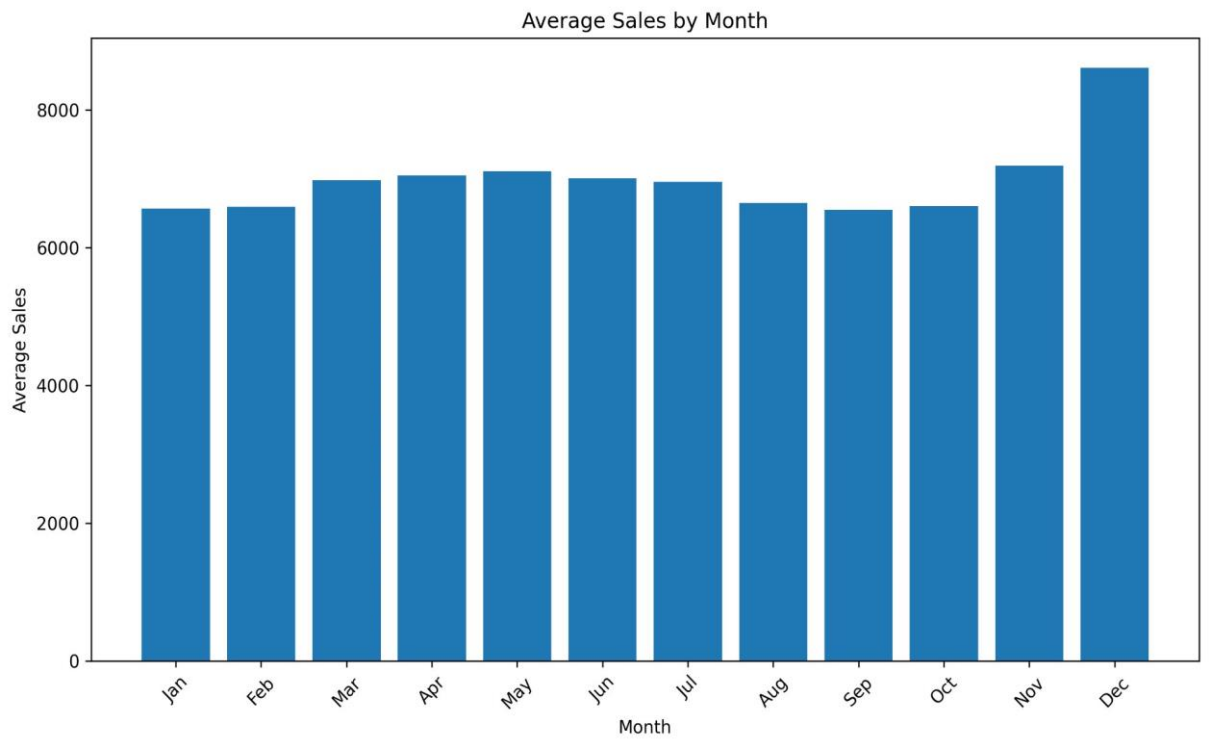
train.csv (Sales History) → store.csv (Store Metadata) → Merge on Store ID → Feature Engineering → Filter Open & Sales > 0 → Train / Test Split (90/10)

KEY MODELLING DECISIONS

✓ **Target variable: Sales**	*Customers excluded, not observable at forecast origin (avoids future data leakage)*
✓ **Temporal train/test split (90/10)**	*Random splits contaminate the test set with future info, invalidated for time series*
✓ **Filter Open=0 and Sales≤0**	*Closed-day sales are deterministic business rules, not stochastic forecasting targets*

Monthly Sales Seasonality

Conditional on open days with positive sales • Average across all stores 2013–2015



€8,609

December Peak

Highest avg. daily sales

+30%

Dec vs Oct

Uplift vs weakest month

Nov

Pre-Xmas Lift

Second strongest month

💡 Business implication: Q4 inventory build-up, peak staffing in Nov–Dec, and tighter cash-flow forecasting in Q4 are directly supported by this pattern.

Day-of-Week Sales Patterns & Sunday Closure Effect

Warning: Sunday averages are conditional on the rare open Sunday — not representative of all Sundays

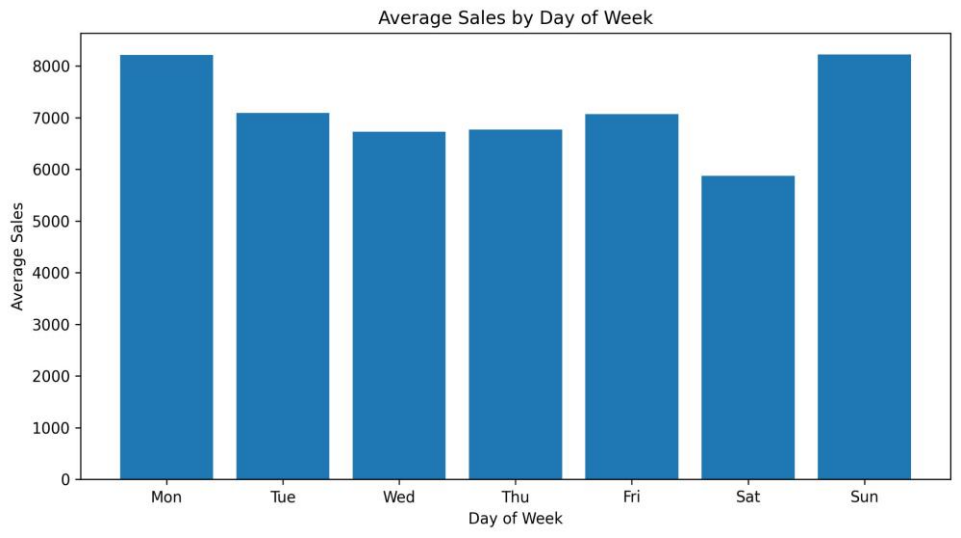


Chart A – Average Sales by Day (open days only)

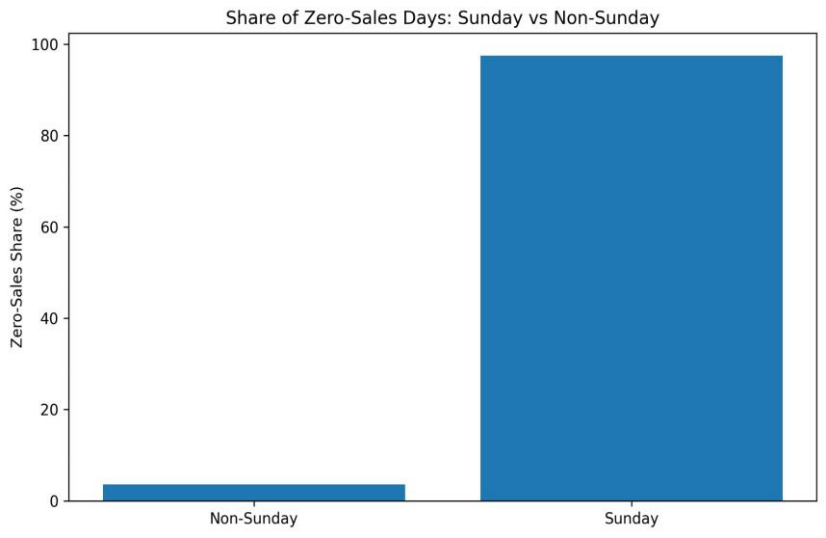
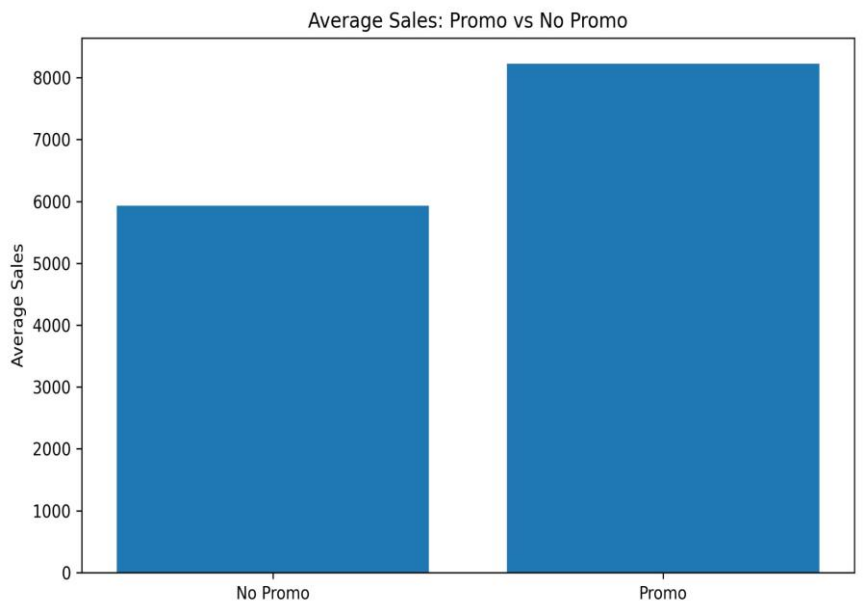


Chart B – Share of Zero-Sales Days: Sunday vs Non-Sunday

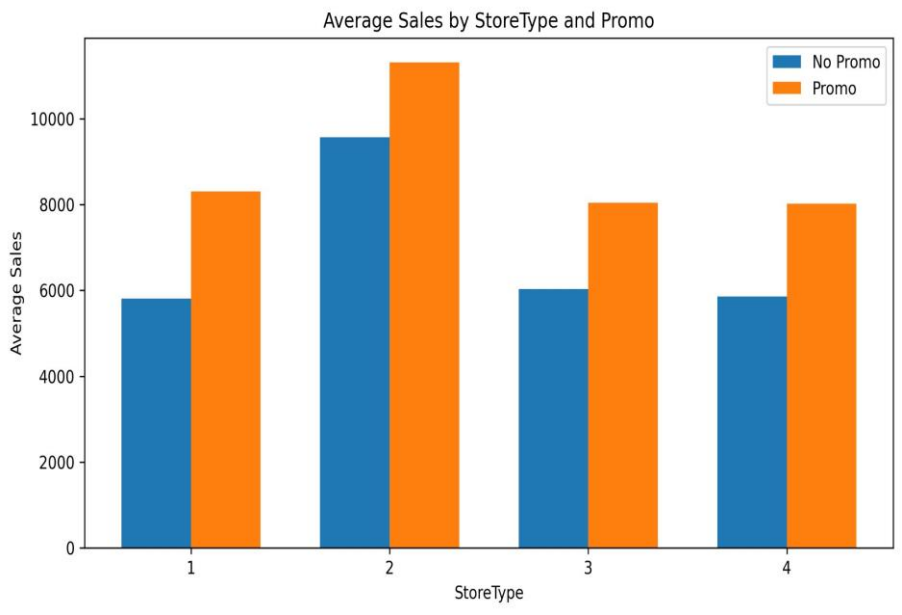
Monday / Sunday high average	*Exceptional open Sundays are high-volume but rare. ~97% of Sundays record zero sales.*
Saturday is weakest weekday	*Structural midweek trough. Saturday historically lower than Mon–Fri average.*
Two-stage forecast logic	*Production system should model P(store open | Sunday) separately from sales level.*

Promotional Uplift: The Single Strongest Commercial Lever

All store types respond to Promo, but response intensity varies by format



Aggregate Promo Effect



Promo × Store Type Interaction

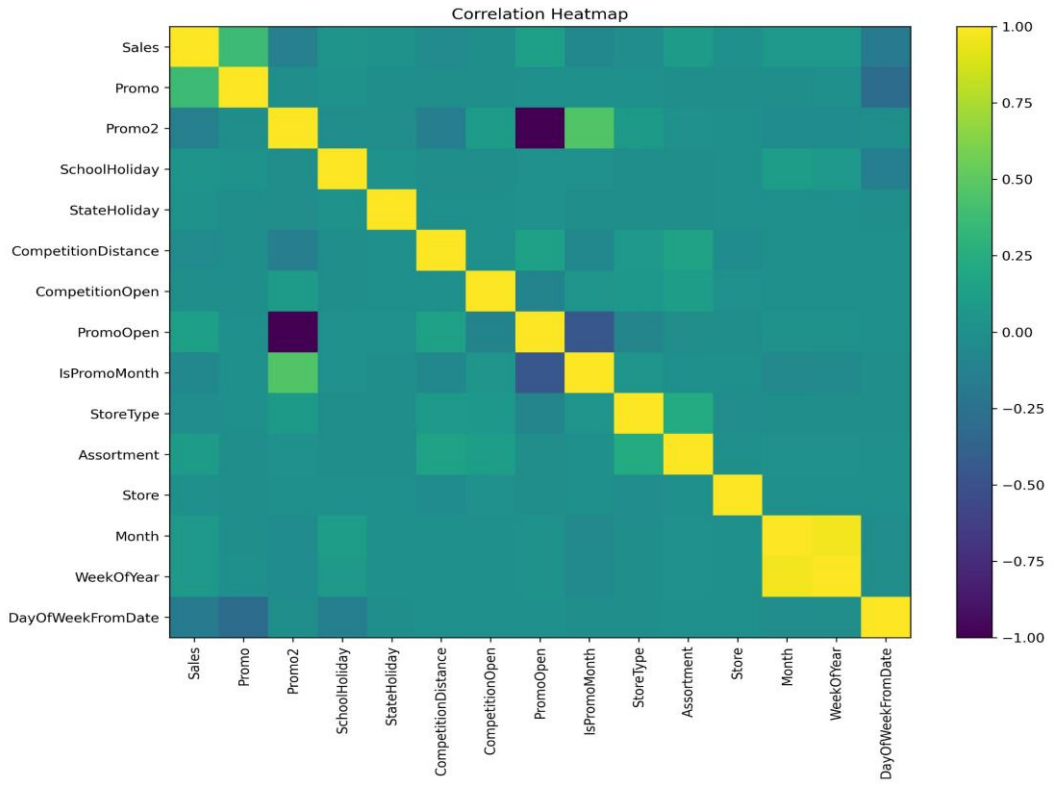
+38.8%
Average Promo Uplift
Promo vs No-Promo days

Type 2
Best Promo Responder
Highest absolute uplift

Promo
#1 XGBoost Feature
By model importance score

Variable Correlation Structure

Useful for feature screening — note: nonlinear relationships not captured here



Promo → Sales (+)

Strongest linear signal in the matrix. Promo activation meaningfully correlates with higher daily sales.

DayOfWeek → Sales (−)

Later-numbered days (Sat=6) are associated with lower sales, consistent with the EDA.

Month ↔ WeekOfYear

Near-perfect collinearity, only one should typically be used in linear models. Trees handle both.

PromoOpen ↔ IsPromoMonth (−)

Negative correlation is by construction: PromoOpen measures elapsed duration, IsPromoMonth flags the window.

StoreType encoding caveat

Integer encoding imposes false ordinal scale (a=1 < b=2). Works for XGBoost; misleading in linear models.

Forecasting Framework & Model Selection

Three models compared: Naive Baseline · Linear Regression · XGBoost Regressor

BENCHMARK
Naive Baseline

- Store × Day-of-Week historical mean
- Falls back to store mean, then global mean
- No parameters, no training overhead
- Surprisingly competitive on RMSPE

REFERENCE
Linear Regression

- OLS on all engineered features
- Integer-encoded categoricals
- Assumes additive linear relationships
- Fails to capture nonlinearities & interactions

CHAMPION
XGBoost

- n_estimators=100, max_depth=8
- learning_rate=0.1, subsample=0.8
- colsample_bytree=0.8
- Best MAE —> captures nonlinear promo effects

TEMPORAL VALIDATION DESIGN

TRAINING SET
Jan 2013 – Apr 2015
(90% of data)

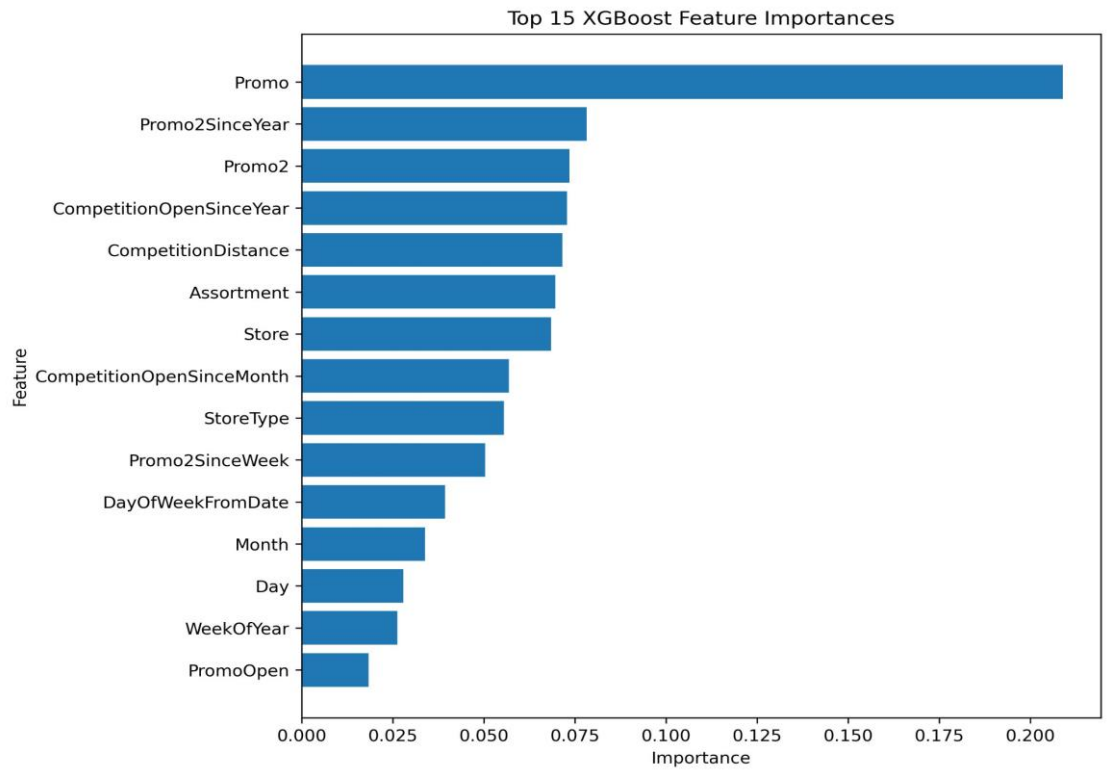
TEST SET
May – Jul 2015
(10% of data)

NO DATA LEAKAGE
Time-ordered sort
before split

DUAL METRIC
Evaluate on RMSPE
and MAE

XGBoost Feature Importances: What Drives Sales ?

Gain-based importance. Note: directional interpretation requires SHAP (future work)



TIER 1 — DOMINANT

Promo

By far the strongest signal. 3× more important than the next feature. Promotions are the primary controllable lever.

TIER 2 — STRUCTURAL

Promo2SinceYear, Promo2, CompetitionOpenSinceYear, CompetitionDistance, Assortment, Store

Store identity and promotional programme design. Capture cross-sectional heterogeneity.

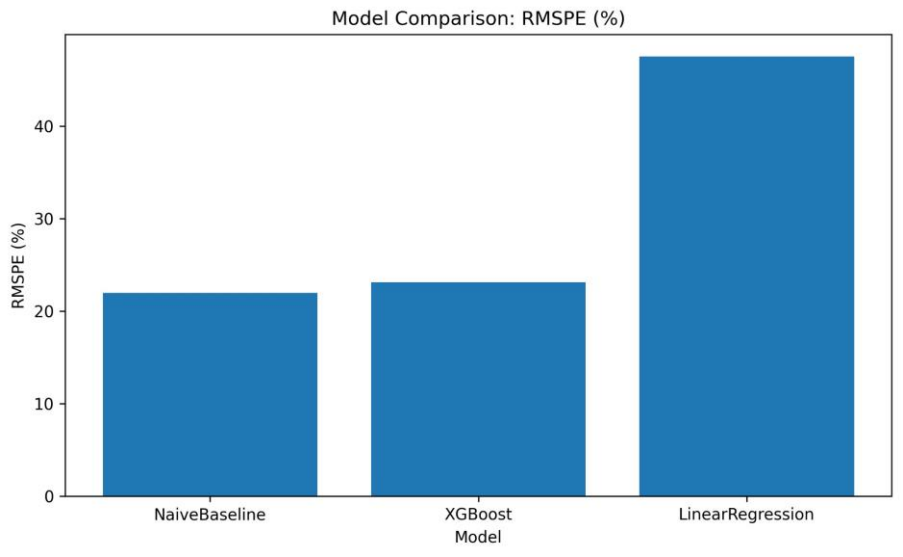
TIER 3 — CALENDAR

DayOfWeekFromDate, Month, Day, WeekOfYear, PromoOpen

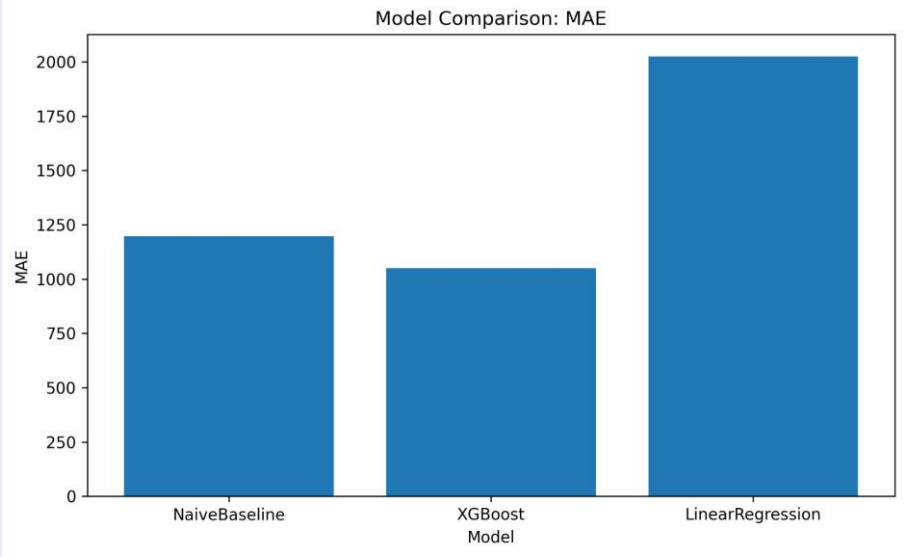
Temporal rhythm. Important but secondary to the structural and promo signals.

Model Performance: The Metric Divergence Story

XGBoost wins on MAE (€ error) but the Naive Baseline wins on RMSPE — both results matter



RMSPE (%) — lower is better

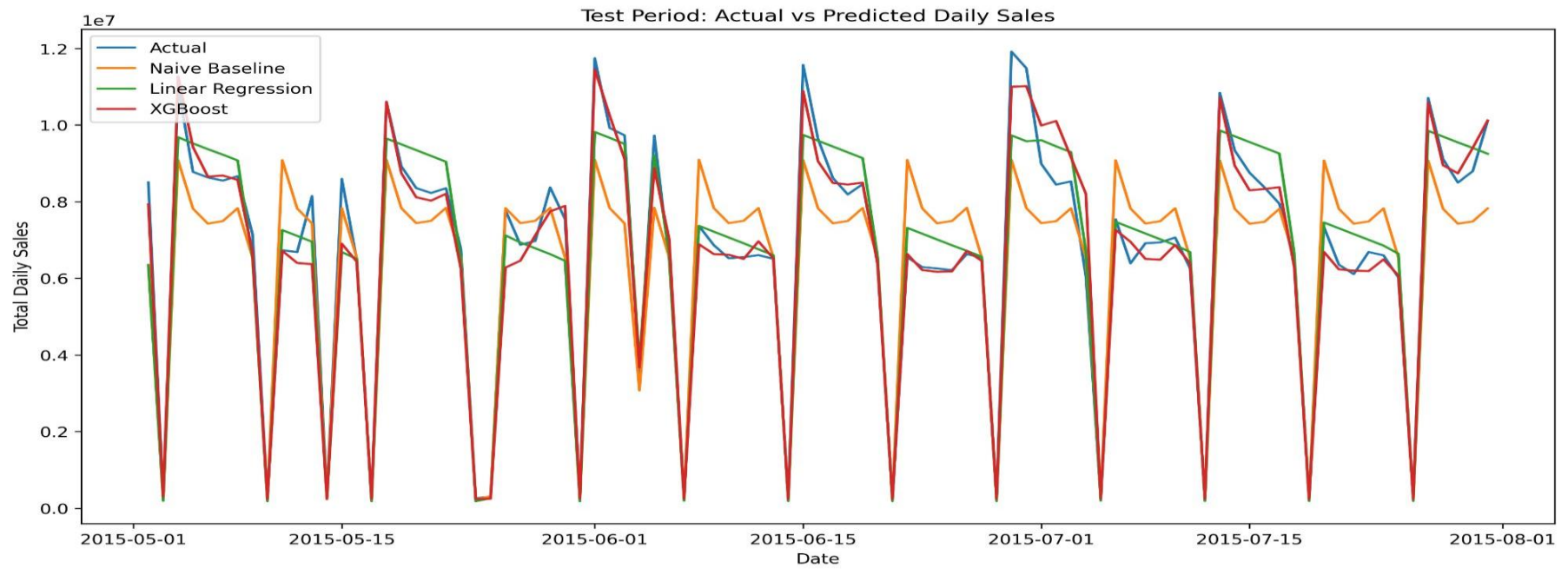


MAE (€) — lower is better

Model	RMSPE (%)	MAE (€)	RMSE (€)	Winner?
NaiveBaseline	21.98%	~1,200	~1,900	✓ RMSPE
XGBoost	23.0%	~1,051	~1,700	✓ MAE / RMSE
LinearRegression	47%+	~2,030	~2,800	✗ Weakest

Aggregate Test Period: Actual vs Predicted Daily Sales

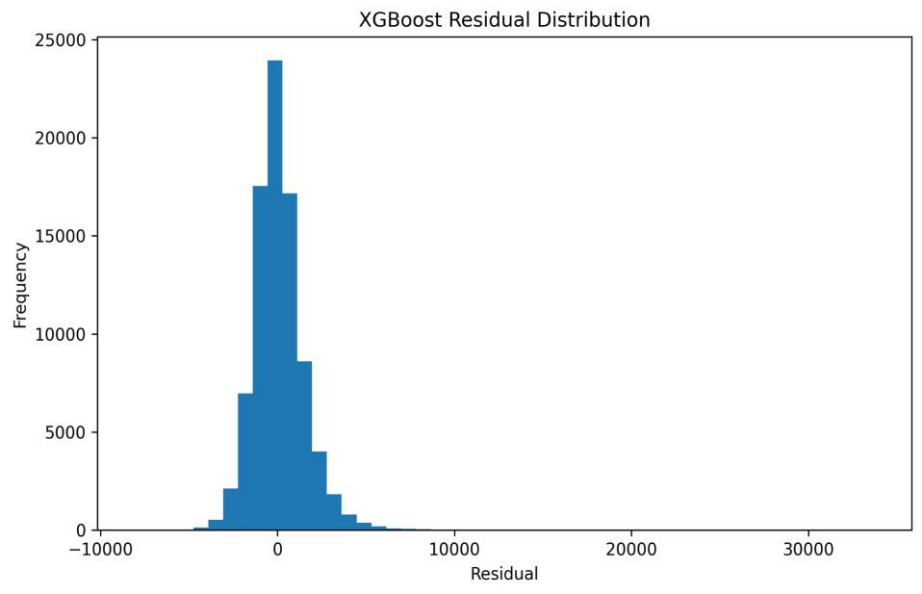
May – July 2015 • All stores summed • XGBoost tracks rhythm; Linear Regression too smooth



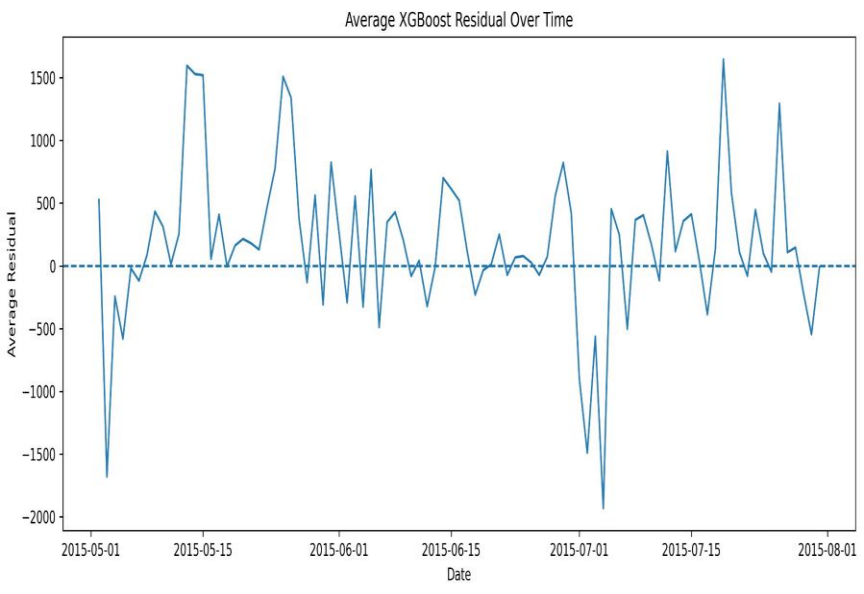
- **XGBoost (Red):** *Tracks weekly rhythm and major spikes well. Collapses correctly on Sunday closures.*
- **Linear Regression (Green):** *Too smooth, misses the amplitude of weekly swings. RMSPE confirms the failure.*
- **Naive Baseline (Orange):** *Captures periodic structure, less responsive to promotional peaks vs XGBoost.*
- **Aggregation caveat:** *Chain-level fit can look clean while store-level errors cancel out. See store-level diagnostics.*

Residual Structure: Model Errors Are Not Random

Right-skewed residuals and time-varying bias indicate unmodelled demand drivers



Residual Distribution (Actual − Predicted)

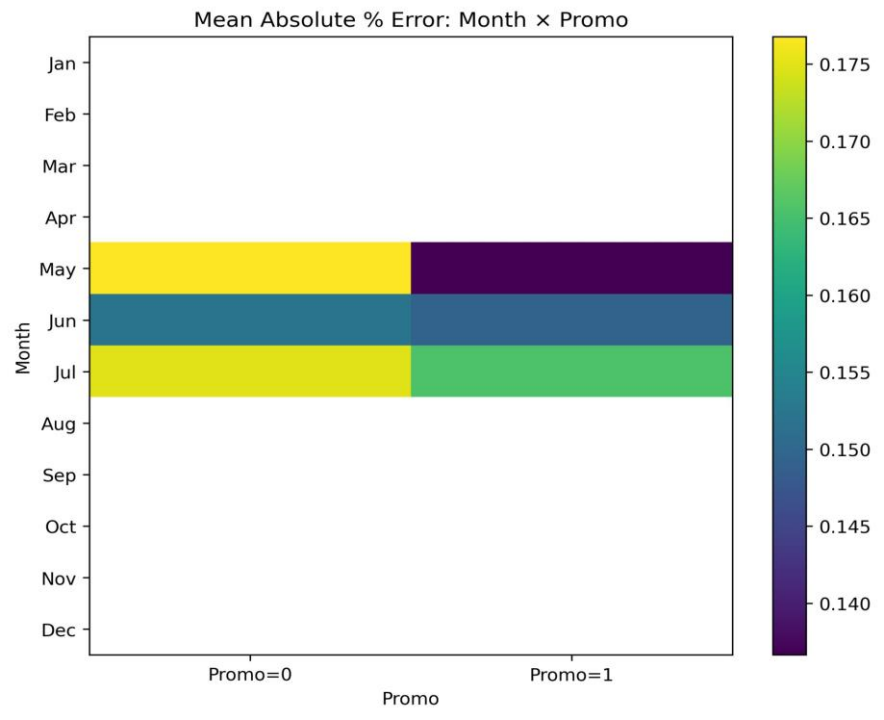


Average Residual Over Time

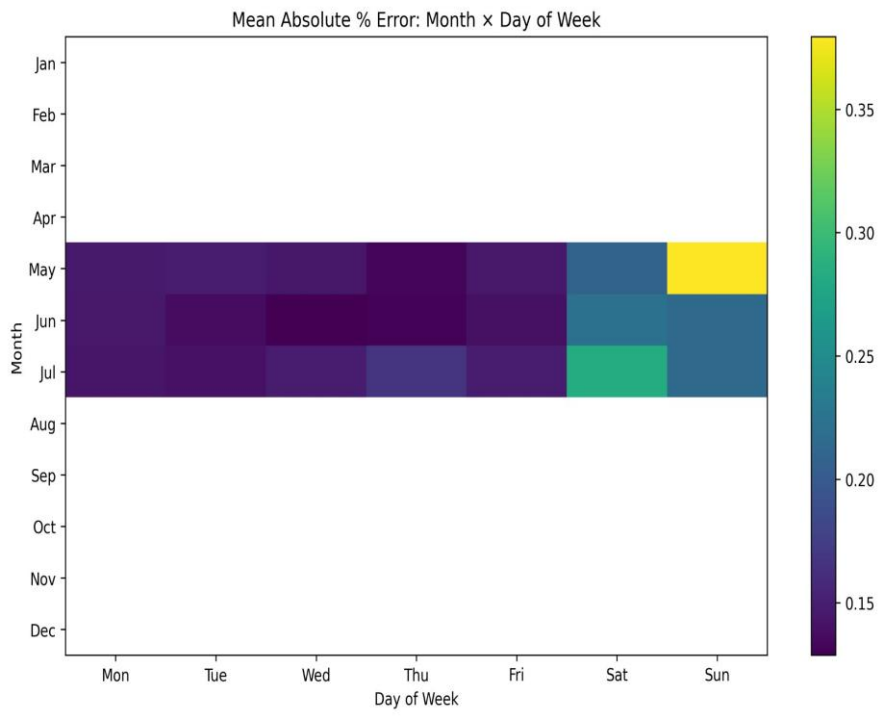
Right tail → underprediction of peaks	*The model is biased in the upside tail. Stockouts during promotional spikes are the key operational risk.*
Time-varying bias → non-stationary model	*Large negative residuals in early July suggest unmodelled event or regime. Model should be monitored in production.*
Center near-zero → adequate baseline	*For typical trading days the model is unbiased. The challenge is at the extremes, not the average.*

Error Heatmaps: Where the Model Fails

Test period covers May–July 2015 only — blank months are not model failure, they are out-of-test



MAPE by Month × Promo Status



MAPE by Month × Day of Week

From Research to Operations: What This Means

Translating model findings into actionable operational decisions

📦 Inventory & Supply Chain

› Build seasonal stock positions from October — December demand is 30% above the annual average

› Promo periods require pre-positioned inventory — demand uplift is ~39% and model confirms it is the #1 driver

› Weekend replenishment requires wider safety buffers: MAPE on Sunday is structurally highest

👥 Labor Planning

› Align staffing calendars to the Month × Day-of-Week heatmap — nonlinear interactions require granular scheduling

› December Monday peaks need maximum staffing; October Saturday needs minimum

› Promo activation signals are known in advance — use them to drive staffing uplift weeks ahead

💰 Finance & Revenue Planning

› XGBoost provides ~€1,051 MAE at store level — suitable for bottom-up revenue aggregation

› Positive right-tail residuals mean revenue upside is more frequently missed than downside: plan conservatively

› Store 1114-type systematic underprediction: flag high-volume stores for manual override review

📣 Promotion Strategy

› Promo is the single largest controllable lever — 38.8% average uplift confirmed

› Store Type 2 has the highest absolute promo response — prioritise promo allocation to this format

› Model errors are lower on Promo=1 days in May — structured demand is more predictable than unstructured

Roadmap: From Prototype to Production

Five prioritised improvements with estimated impact on forecast error

01

SHAP Value Analysis

Immediate | **High IMPACT**

Replace gain-based importance with SHAP. Enables directional interpretation and identifies which features push individual predictions up or down.

02

Rolling Window Backtesting

2–4 Weeks | **High IMPACT**

Validate on multiple test windows across seasons, not just one holdout. Stress-test model on holiday periods and promo regimes not seen in training.

03

Two-Stage Closure Model

4–8 Weeks | **Medium IMPACT**

Add a binary open/closed classifier as Stage 1. Feed its output to Stage 2 (sales level). Critical for Sunday and public holiday prediction.

04

Store-Level Hypertuning

6–10 Weeks | **Medium IMPACT**

High-MAE stores (e.g. Store 1114) warrant dedicated model calibration or ensemble blending with a local naive baseline.

05

Causal Promo Estimation

Ongoing | **High IMPACT**

Move from associational promo uplift to causal estimates using Diff-in-Diff or synthetic control. Required for ROI calculation and budget allocation.

KEY TAKEAWAYS

What This Research Proves

- Promotions are the #1 controllable commercial lever — 38.8% average uplift, confirmed across all store types
- December demand is 30%+ above the annual average — Q4 planning is the highest-stakes operational window
- XGBoost achieves €1,051 MAE and is suitable for bottom-up revenue aggregation at chain level
- The model has a structural right-tail gap — peaks are underforecast — operational buffers remain essential
- Naive Baseline outperforms XGBoost on RMSPE: a disciplined evaluation requires both metrics
- A two-stage closure + sales model and SHAP analysis are the clearest priority improvements

Metrics Summary

~23%
XGBoost RMSPE

€1,051
XGBoost MAE

~22%
Naive RMSPE

+38.8%
Promo Uplift